ASSIGNMENT 1 - PROJECTION

Qs1: WRITE A QUERY TO DISPLAY ALL THE DETAILS FROM THE EMPLOYEE TABLE.

```
mysql> CREATE TABLE EMP (
    ->     EMPNO INT PRIMARY KEY,
    ->     ENAME VARCHAR(20),
    ->     JOB VARCHAR(20),
    ->     SAL DECIMAL(10,2));
Query OK, 0 rows affected (0.05 sec)
```

```
mysql> INSERT INTO EMP VALUES (101, 'RAHUL', 'MANAGER', 5000.00);
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO EMP VALUES (102, 'ANITA', 'CLERK', 2000.00);
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO EMP VALUES (103, 'VIKAS', 'SALESMAN', 3000.00);
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO EMP VALUES (104, 'NEHA', 'ANALYST', 4000.00);
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO EMP VALUES (105, 'ROHIT', 'CLERK', 1800.00);
Query OK, 1 row affected (0.01 sec)

mysql> select * from emp;
+-------+-------+----------+---------+
| EMPNO | ENAME | JOB      | SAL     |
+-------+-------+----------+---------+
|   101 | RAHUL | MANAGER  | 5000.00 |
|   102 | ANITA | CLERK    | 2000.00 |
|   103 | VIKAS | SALESMAN | 3000.00 |
|   104 | NEHA  | ANALYST  | 4000.00 |
|   105 | ROHIT | CLERK    | 1800.00 |
+-------+-------+----------+---------+
5 rows in set (0.00 sec)
```

Qs 2: WAQTD NAMES OF ALL THE EMPLOYEES.

```
mysql> SELECT ENAME FROM EMP;
+-------+
| ENAME |
+-------+
| RAHUL |
| ANITA |
| VIKAS |
| NEHA  |
| ROHIT |
+-------+
5 rows in set (0.00 sec)
```

Qs 3: 3 WAQTD NAME AND SALARY GIVEN TO ALL THE EMPLOYEES.

```
mysql> SELECT ENAME, SAL FROM EMP;
+-------+---------+
| ENAME | SAL     |
+-------+---------+
| RAHUL | 5000.00 |
| ANITA | 2000.00 |
| VIKAS | 3000.00 |
| NEHA  | 4000.00 |
| ROHIT | 1800.00 |
+-------+---------+
5 rows in set (0.00 sec)
```

Qs 4 WAQTD NAME AND COMMISSION GIVEN TO ALL THE EMPLOYEES.

```
mysql> UPDATE EMP SET COMM = 0 WHERE EMPNO = 102;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> UPDATE EMP SET COMM = 300 WHERE EMPNO = 103;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> UPDATE EMP SET COMM = 0 WHERE EMPNO = 104;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> UPDATE EMP SET COMM = 200 WHERE EMPNO = 105;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT ENAME, COMM FROM EMP;
+-------+--------+
| ENAME | COMM   |
+-------+--------+
| RAHUL | 500.00 |
| ANITA |   0.00 |
| VIKAS | 300.00 |
| NEHA  |   0.00 |
| ROHIT | 200.00 |
+-------+--------+
5 rows in set (0.00 sec)
```

Qs 5 WAQTD EMPLOYEE ID AND DEPARTMENT NUMBER OF ALL THE EMPLOYEES IN EMP TABLE.

```
mysql> ALTER TABLE EMP ADD DEPTNO INT;
Query OK, 0 rows affected (0.04 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> UPDATE EMP SET DEPTNO = 10 WHERE EMPNO = 101;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> UPDATE EMP SET DEPTNO = 20 WHERE EMPNO = 102;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> UPDATE EMP SET DEPTNO = 30 WHERE EMPNO = 103;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> UPDATE EMP SET DEPTNO = 20 WHERE EMPNO = 104;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> UPDATE EMP SET DEPTNO = 30 WHERE EMPNO = 105;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT EMPNO, DEPTNO FROM EMP;
+-------+--------+
| EMPNO | DEPTNO |
+-------+--------+
|   101 |     10 |
|   102 |     20 |
|   103 |     30 |
|   104 |     20 |
|   105 |     30 |
+-------+--------+
5 rows in set (0.00 sec)

mysql>
```

Qs 6 WAQTD ENAME AND HIREDATE OF ALL THE EMPLOYEES.

```
mysql> UPDATE EMP SET HIREDATE = '2022-07-19' WHERE EMPNO = 103;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> UPDATE EMP SET HIREDATE = '2019-11-25' WHERE EMPNO = 104;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> UPDATE EMP SET HIREDATE = '2023-05-12' WHERE EMPNO = 105;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT ENAME, HIREDATE FROM EMP;
+-------+------------+
| ENAME | HIREDATE   |
+-------+------------+
| RAHUL | 2020-01-15 |
| ANITA | 2021-03-10 |
| VIKAS | 2022-07-19 |
| NEHA  | 2019-11-25 |
| ROHIT | 2023-05-12 |
+-------+------------+
5 rows in set (0.00 sec)
```

Qs 7 WAQTD NAME AND DESIGNATION OF ALL THE EMPLOYEES.

```
mysql> SELECT ENAME, JOB FROM EMP;
+-------+----------+
| ENAME | JOB      |
+-------+----------+
| RAHUL | MANAGER  |
| ANITA | CLERK    |
| VIKAS | SALESMAN |
| NEHA  | ANALYST  |
| ROHIT | CLERK    |
+-------+----------+
5 rows in set (0.00 sec)
```

Qs 8 WAQTD NAME, JOB AND SALARY GIVEN TO ALL THE EMPLOYEES.

```
mysql> SELECT ENAME, JOB, SAL FROM EMP;
+-------+----------+---------+
| ENAME | JOB      | SAL     |
+-------+----------+---------+
| RAHUL | MANAGER  | 5000.00 |
| ANITA | CLERK    | 2000.00 |
| VIKAS | SALESMAN | 3000.00 |
| NEHA  | ANALYST  | 4000.00 |
| ROHIT | CLERK    | 1800.00 |
+-------+----------+---------+
5 rows in set (0.00 sec)
```

Qs 9 WAQTD DNAMES PRESENT IN DEPARTMENT TABLE.

```
mysql> CREATE TABLE DEPT (
    ->     DEPTNO INT PRIMARY KEY,
    ->     DNAME VARCHAR(20),
    ->     LOC VARCHAR(20)
    -> );
Query OK, 0 rows affected (0.05 sec)

mysql> INSERT INTO DEPT VALUES (10, 'ACCOUNTING', 'NEW YORK');
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO DEPT VALUES (20, 'RESEARCH', 'DALLAS');
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO DEPT VALUES (30, 'SALES', 'CHICAGO');
Query OK, 1 row affected (0.01 sec)

mysql> SELECT DNAME FROM DEPT;
+------------+
| DNAME      |
+------------+
| ACCOUNTING |
| RESEARCH   |
| SALES      |
+------------+
3 rows in set (0.00 sec)
```

Qs 10 WAQTD DNAME AND LOCATION PRESENT IN DEPT TABLE.

```
mysql> SELECT DNAME, LOC FROM DEPT;
+------------+----------+
| DNAME      | LOC      |
+------------+----------+
| ACCOUNTING | NEW YORK |
| RESEARCH   | DALLAS   |
| SALES      | CHICAGO  |
+------------+----------+
3 rows in set (0.00 sec)

mysql>
```

ASSIGNMENT 1 – ALIAS

1 WAQTD NAME OF THE EMPLOYEE ALONG WITH THEIR ANNUAL SALARY.

```
mysql> SELECT ENAME, SAL * 12 AS ANNUAL_SALARY
    -> FROM EMP;
+-------+---------------+
| ENAME | ANNUAL_SALARY |
+-------+---------------+
| RAHUL |      60000.00 |
| ANITA |      24000.00 |
| VIKAS |      36000.00 |
| NEHA  |      48000.00 |
| ROHIT |      21600.00 |
+-------+---------------+
5 rows in set (0.01 sec)
```

2 WAQTD ENAME AND JOB FOR ALL THE EMPLOYEE WITH THEIR HALF TERM SALARY.

```
mysql> SELECT ENAME, JOB, SAL / 2 AS HALF_TERM_SALARY
    -> FROM EMP;
+-------+----------+------------------+
| ENAME | JOB      | HALF_TERM_SALARY |
+-------+----------+------------------+
| RAHUL | MANAGER  |      2500.000000 |
| ANITA | CLERK    |      1000.000000 |
| VIKAS | SALESMAN |      1500.000000 |
| NEHA  | ANALYST  |      2000.000000 |
| ROHIT | CLERK    |       900.000000 |
+-------+----------+------------------+
5 rows in set (0.00 sec)
```

3 WAQTD ALL THE DETAILS OF THE EMPLOYEES ALONG WITH AN ANNUAL BONUS OF 2000.

```
mysql> SELECT EMP.*, 2000 AS ANNUAL_BONUS
    -> FROM EMP;
+-------+-------+----------+---------+--------+--------+------------+--------------+
| EMPNO | ENAME | JOB      | SAL     | COMM   | DEPTNO | HIREDATE   | ANNUAL_BONUS |
+-------+-------+----------+---------+--------+--------+------------+--------------+
|   101 | RAHUL | MANAGER  | 5000.00 | 500.00 |     10 | 2020-01-15 |         2000 |
|   102 | ANITA | CLERK    | 2000.00 |   0.00 |     20 | 2021-03-10 |         2000 |
|   103 | VIKAS | SALESMAN | 3000.00 | 300.00 |     30 | 2022-07-19 |         2000 |
|   104 | NEHA  | ANALYST  | 4000.00 |   0.00 |     20 | 2019-11-25 |         2000 |
|   105 | ROHIT | CLERK    | 1800.00 | 200.00 |     30 | 2023-05-12 |         2000 |
+-------+-------+----------+---------+--------+--------+------------+--------------+
5 rows in set (0.00 sec)
```

4 WAQTD NAME SALARY AND SALARY WITH A HIKE OF 10%.

```
mysql> SELECT ENAME,
    ->        SAL,
    ->        SAL + (SAL * 0.10) AS SALARY_AFTER_10_PERCENT_HIKE
    -> FROM EMP;
+-------+---------+-----------------------------+
| ENAME | SAL     | SALARY_AFTER_10_PERCENT_HIKE |
+-------+---------+-----------------------------+
| RAHUL | 5000.00 |                   5500.0000 |
| ANITA | 2000.00 |                   2200.0000 |
| VIKAS | 3000.00 |                   3300.0000 |
| NEHA  | 4000.00 |                   4400.0000 |
| ROHIT | 1800.00 |                   1980.0000 |
+-------+---------+-----------------------------+
5 rows in set (0.00 sec)
```

5 WAQTD NAME AND SALARY WITH DEDUCTION OF 25%.

```
mysql> SELECT ENAME,
    ->        SAL,
    ->        SAL - (SAL * 0.25) AS SALARY_AFTER_25_PERCENT_DEDUCTION
    -> FROM EMP;
+-------+---------+----------------------------------+
| ENAME | SAL     | SALARY_AFTER_25_PERCENT_DEDUCTION |
+-------+---------+----------------------------------+
| RAHUL | 5000.00 |                        3750.0000 |
| ANITA | 2000.00 |                        1500.0000 |
| VIKAS | 3000.00 |                        2250.0000 |
| NEHA  | 4000.00 |                        3000.0000 |
| ROHIT | 1800.00 |                        1350.0000 |
+-------+---------+----------------------------------+
5 rows in set (0.01 sec)
```

6 WAQTD NAME AND SALARY WITH MONTHLY HIKE OF 50.

```
mysql> SELECT ENAME,
    ->        SAL,
    ->        SAL + 50 AS SALARY_AFTER_MONTHLY_HIKE
    -> FROM EMP;
+-------+---------+---------------------------+
| ENAME | SAL     | SALARY_AFTER_MONTHLY_HIKE |
+-------+---------+---------------------------+
| RAHUL | 5000.00 |                   5050.00 |
| ANITA | 2000.00 |                   2050.00 |
| VIKAS | 3000.00 |                   3050.00 |
| NEHA  | 4000.00 |                   4050.00 |
| ROHIT | 1800.00 |                   1850.00 |
+-------+---------+---------------------------+
5 rows in set (0.00 sec)
```

7 WAQTD NAME AND ANNUAL SALARY WITH DEDUCTION OF 10%.

```
mysql> SELECT ENAME,
    ->        SAL * 12 AS ANNUAL_SALARY,
    ->        (SAL * 12) - (SAL * 12 * 0.10) AS ANNUAL_SALARY_AFTER_10_PERCENT_DEDUCTION
    -> FROM EMP;
+-------+---------------+-----------------------------------------+
| ENAME | ANNUAL_SALARY | ANNUAL_SALARY_AFTER_10_PERCENT_DEDUCTION |
+-------+---------------+-----------------------------------------+
| RAHUL |      60000.00 |                              54000.0000 |
| ANITA |      24000.00 |                              21600.0000 |
| VIKAS |      36000.00 |                              32400.0000 |
| NEHA  |      48000.00 |                              43200.0000 |
| ROHIT |      21600.00 |                              19440.0000 |
+-------+---------------+-----------------------------------------+
5 rows in set (0.00 sec)
```

8 WAQTD TOTAL SALARY GIVEN TO EACH EMPLOYEE (SAL+COMM).

```
mysql> SELECT ENAME,
    ->        SAL,
    ->        COMM,
    ->        SAL + COMM AS TOTAL_SALARY
    -> FROM EMP;
+-------+---------+--------+--------------+
| ENAME | SAL     | COMM   | TOTAL_SALARY |
+-------+---------+--------+--------------+
| RAHUL | 5000.00 | 500.00 |      5500.00 |
| ANITA | 2000.00 |   0.00 |      2000.00 |
| VIKAS | 3000.00 | 300.00 |      3300.00 |
| NEHA  | 4000.00 |   0.00 |      4000.00 |
| ROHIT | 1800.00 | 200.00 |      2000.00 |
+-------+---------+--------+--------------+
5 rows in set (0.00 sec)
```

9 WAQTD DETAILS OF ALL THE EMPLOYEES ALONG WITH ANNUAL SALARY.

```
mysql> SELECT EMP.*,
    ->        SAL * 12 AS ANNUAL_SALARY
    -> FROM EMP;
+-------+-------+----------+---------+--------+--------+------------+---------------+
| EMPNO | ENAME | JOB      | SAL     | COMM   | DEPTNO | HIREDATE   | ANNUAL_SALARY |
+-------+-------+----------+---------+--------+--------+------------+---------------+
|   101 | RAHUL | MANAGER  | 5000.00 | 500.00 |     10 | 2020-01-15 |      60000.00 |
|   102 | ANITA | CLERK    | 2000.00 |   0.00 |     20 | 2021-03-10 |      24000.00 |
|   103 | VIKAS | SALESMAN | 3000.00 | 300.00 |     30 | 2022-07-19 |      36000.00 |
|   104 | NEHA  | ANALYST  | 4000.00 |   0.00 |     20 | 2019-11-25 |      48000.00 |
|   105 | ROHIT | CLERK    | 1800.00 | 200.00 |     30 | 2023-05-12 |      21600.00 |
+-------+-------+----------+---------+--------+--------+------------+---------------+
5 rows in set (0.00 sec)
```

10 WAQTD NAME AND DESIGNATION ALONG WITH 100 PENALTY IN SALARY.

```
mysql> SELECT ENAME,
    ->        JOB,
    ->        SAL,
    ->        SAL - 100 AS SALARY_AFTER_100_PENALTY
    -> FROM EMP;
+-------+----------+---------+-------------------------+
| ENAME | JOB      | SAL     | SALARY_AFTER_100_PENALTY |
+-------+----------+---------+-------------------------+
| RAHUL | MANAGER  | 5000.00 |                  4900.00 |
| ANITA | CLERK    | 2000.00 |                  1900.00 |
| VIKAS | SALESMAN | 3000.00 |                  2900.00 |
| NEHA  | ANALYST  | 4000.00 |                  3900.00 |
| ROHIT | CLERK    | 1800.00 |                  1700.00 |
+-------+----------+---------+-------------------------+
5 rows in set (0.00 sec)

mysql>
```